Module Code: **GENMP402**

Module Name: Applied Physics/Apply Mechanics and Properties of Matter

Date: **22/June/2026**

Period: **8:30 AM- 11:30 AM**

NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

END OF TERM III EXAMINATION, 2025-2026

QUESTION PAPER

RQF LEVEL : 4

TRADES : AUT, WIR, FOR, WOT, NIT, SWD, CSA, ETE, ELT, REN, LSV, PLT, IND, BDC, PWO, MAT & MMP.

DURATION : 3 HOURS

MARKS : /100 **CAMIS** /30

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DURATION : 3 HOURS

INSTRUCTIONS:

1) This paper consists of three sections: **A, B** and **C**.
   - **Section A**: Attempt all questions. **(55 marks)**
   - **Section B**: Attempt any three questions. **(30 marks)**
   - **Section C**: Compulsory question **(15 marks)**
2) Use only a **blue** or **black** pen for answering and a **pencil** for drawing.
3) For Multiple Choice Questions, **choose** a letter corresponding to the correct answer.
4) The marks to be recorded into CAMIS will result from the conversion of **100** into **30**

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SECTION A: ATTEMPT ALL QUESTIONS (55 marks)

1) In relation to the effect of forces

I) A force can change _____.

A) the size or shape of a body
B) the state of rest or of motion of a body
C) the dimensions of a body
D) all the above **(1 mark)**

II) The non-contact force is _____.

A) force of reaction
B) force due to gravity
C) tension in string
D) force of the friction **(1 mark)**

III) When a comb is rubbed on dry hair, it gets charged. This is an example of_____.

A) Gravitational force
B) normal reaction force
C) magnetic force
D) electrostatic force. **(1 mark)**

2) In relation to the quantization of the Newtons gravitation force and Newtons laws of motion:

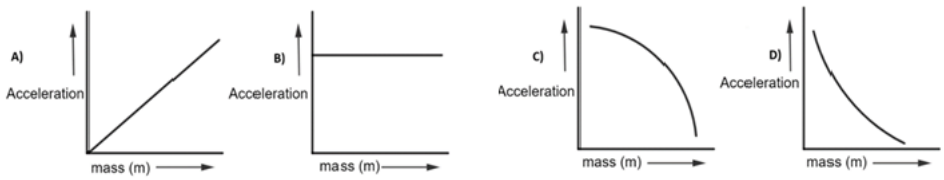
I) On doubling the separation between two bodies, the magnitude of the non-contact force between them becomes _____.

A) one-eighth
B) one-half
C) one-fourth
D) remains the same **(1 mark)**

II) The force required to produce an acceleration of 5 m s^{-2} in a body of mass 2 kg is _____.

A) 2.5 N
B) 20 N
C) 10 N
D) 25 N **(2 marks)**

III) For a given force **F**, the graph plotted for acceleration against mass is: **(1 mark)**



3) Complete the sentences using the following words: **static, dynamic, zero**, **(Nm)** as related to equilibrium of bodies.

I) The S.I. unit of moment of force is _____. **(1 mark)**

II) In equilibrium algebraic sum of moments of all forces about the point of rotation is _____. **(1 mark)**

III) In a beam balance when the beam is balanced in a horizontal position, it is in _____ equilibrium. **(1 mark)**

IV) The moon revolving around the earth is in _____ equilibrium. **(1 mark)**

4) An object is deformed when contact forces are applied to it. A deformation is a

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change in the size or shape of the object. Many solids are stiff enough that the deformation cannot be seen with the human eye; a microscope or other sensitive device is required to detect the change in size or shape. When the contact forces are removed, an elastic object returns to its original shape and size. Many objects are elastic as long as the deforming forces are not too large. On the other hand, any object may be permanently deformed or even broken if the forces acting is too large.

I) A stiff material is characterized by _____.
   A) high ultimate strength.
   B) high breaking strength.
   C) high young's modulus.
   D) high proportional limit. **(1 mark)**

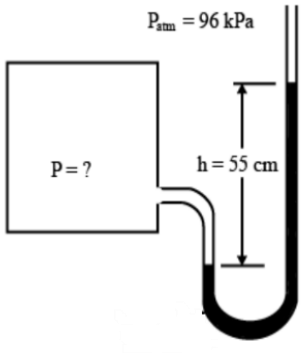
II) A brittle material is characterized by _____.
   A) high breaking strength and low young's modulus.
   B) low breaking strength and high young's modulus.
   C) high breaking strength and high young's modulus.
   D) low breaking strength and low young's modulus. **(1 mark)**

III) Which pair of quantities can be expressed in the same units?
   A) stress and strain
   B) Young's modulus and strain
   C) Young's modulus and stress
   D) ultimate strength and strain **(1 mark)**

5) In relation to atmospheric pressure and variation of pressure in static fluid answer the following questions:

I) On the moon an astronaut cannot drink Fanta with the help straw because:
   A) Acceleration due to gravity on the moon is less
   B) There is no atmosphere on the moon
   C) Fanta evaporate instantaneously on the moon
   D) None of the above **(1 mark)**

II) A manometer is used to measure the pressure of a gas in a tank. The density of the fluid used is 850 kg/m^3 and the manometer column height is 55 cm, as shown in figure. If the local atmospheric pressure is 96 kPa, what is the absolute pressure in the tank?



   A) 4.6 kPa
   B) 98.6 kPa
   C) 100.6 kPa
   D) 200 kPa **(2 marks)**

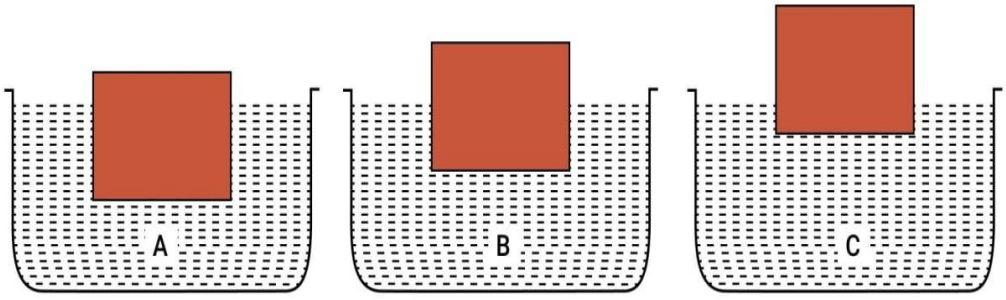
6) Archimedes' principle states that when a body is immersed partially or completely in a liquid, it experiences an upthrust, which is equal to the weight of the liquid displaced by it.

I) The apparent loss in weight is equal to the upthrust on the body verifies:

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A) Pascal's law
B) Archimedes' principle
C) Newton's third law
D) Newton's second law **(1 mark)**

II) The figure below shows the same block of wood floating in three different liquids **A, B** and **C** of densities ρ_1, ρ_2 and ρ_3 respectively. Which liquid has the highest density? Give reason for your answer. **(1 mark)**



III) An unloaded ship is filled with sand at its bottom to _____ its centre of gravity and make its equilibrium _____.
A) lower, unstable
B) higher, stable
C) higher, unstable
D) lower, stable **(2 marks)**

7) Optical instruments help in the magnification of images. Answer the following questions as related to optical instruments.

I) Magnifying power of a simple microscope
A) Increases with increase in focal length
B) Increases with the decrease in focal length
C) No effect with decrease or increase with the focal length
D) Least distance of distinct vision **(1 mark)**

II) The magnifying power of a telescope is 9. When it is adjusted normally, the distance between the objective and the eye-piece is found to be 20 cm. The focal lengths of the lenses are _____.
A) 18 cm, 2 cm
B) 11 cm, 9 cm
C) 10 cm, 10 cm
D) 15 cm, 5 cm. **(2 marks)**

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8) Here are common questions regarding Kirchhoff's Laws: answer them by **True/False.**

I) Kirchhoff's Voltage Law states that the sum of all voltage drops across resistors and emf's around a closed loop is zero. **(1 mark)**

II) Kirchhoff's junction law is based on the law of conservation of energy. **(1 mark)**

III) Kirchhoff's voltage law is based on the law of conservation of energy. **(1 mark)**

IV) Kirchhoff's laws are useful to solve complex circuits. **(1 mark)**

9) Here both optical lens and vision defects match **column X** and **column Y.** **(4 marks)**

Column X: Optical lens and vision Defects	**Column Y: Their meaning and correction**
I) Myopia (Near-sightedness)	A) Nearby objects are blurry; image forms behind the retina. Corrected by Convex (Converging) lens.

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II) Chromatic Aberration	B) Rays passing through the edge of a lens focus at a different point than rays passing near the center.
III) Hypermetropia (Farsightedness)	C) Distant objects are blurry; image forms in front of the retina. Corrected by Concave (Diverging) lens
IV) Spherical Aberration:	D) Different colors of light focus at different positions, causing colored edges around images.

10) Answer the following questions as related to mechanical energy and its conservation law.

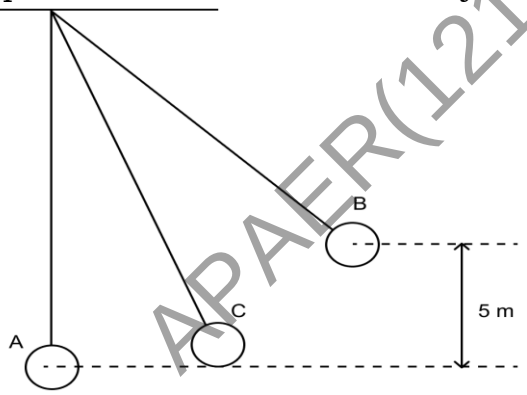
I) Sate two differences between the potential energy and the kinetic energy. **(2 marks)**

II) A body of mass 10 kg falls from a height of 20 m to 8 calculate:

a) the loss in potential energy of the body, and **(2 marks)**

b) the total energy possessed by the body at any instant? (Take g = 10 m s^{-2}) **(2 marks)**

11) The figure below shows a simple pendulum of mass 200 g. It is displaced from the mean position **A** to the extreme position **B**. The potential energy at position **A** is zero. At position **B** the bob is raised by 5 m.



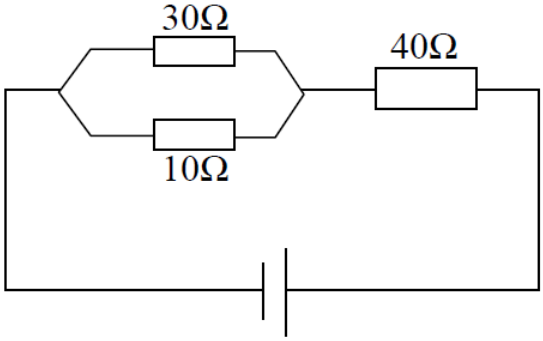
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a) Calculate the potential energy of the pendulum at position **B**? **(2 marks)**

b) Calculate the total mechanical energy at point **C**? **(2 marks)**

c) Calculate the speed of the bob at position A when released from position **B**? (Take g = 10 m s^{-2} **(2 marks)**

12) The diagram in the below figure shows three resistors connected in a circuit. What is the effective resistance of the current?



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A) 20Ω
B) 47.5Ω
C) 60Ω
D) 90Ω **(2 marks)**

13) The gravitational force of attraction between two bodies of masses 60 kg and 40 kg separated by a distance 10 m is: (where G = 6.7 x 10^{-11} N m^2 kg^{-2})
A) 1.60 x 10^{-9} N
B) 2.8 x 10^{-11} N
C) 1.0 x 10^{-11} N
D) 1.6 x 10^{-11} N **(2 marks)**

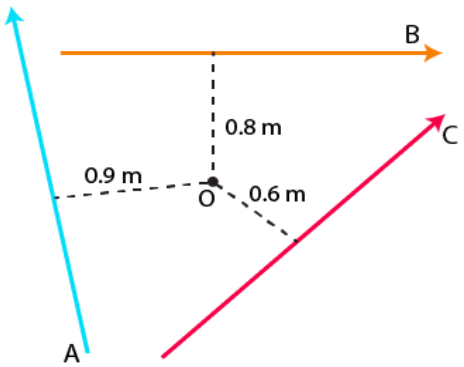
14) Answer the following as related to different forms of energy transformations.
I) In a photoelectric cell, the change in energy is from:
A) electrical to mechanical
B) light energy to electrical energy
C) electrical energy to light energy
D) chemical energy to light energy **(1 mark)**
II) In a petrol vehicle while in motion, the change in energy is from
A) kinetic energy to potential energy
B) chemical energy to kinetic energy
C) potential energy to kinetic energy
D) chemical energy to potential energy **(1 mark)**
III) In nuclear fission:
A) chemical energy changes to electrical energy
B) mechanical energy changes to chemical energy
C) nuclear energy changes to electrical energy
D) electric energy changes to nuclear energy **(1 mark)**

15) Rwanda's energy sector relies on a diverse mix of sources, with a total installed electricity capacity of roughly 408.9 MW (as of 2024), predominantly comprising thermal (51%) and hydropower (43.9%), alongside solar (4.2%). Major sources include Lake Kivu methane gas, imported petroleum, biomass, peat, and emerging solar and geothermal projects. **Give Four ways a student in Rwanda can conserve energy at school.** **(4 marks)**

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SECTION B: ATTEMPT ONLY THREE (3) QUESTIONS (30 marks)

16) **A**, **B** and **C** are the three forces each of magnitude 4 N acting in the plane of the paper as shown in Figure below. The point **O** lies in the same plane.



I) Which force has the least moment about **O**? Give a reason. **(2 marks)**

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II) Which force has the greatest moment about **O**? Give a reason. **(2 marks)**

III) Name the forces producing:

a) clockwise **(2 marks)**

b) anticlockwise moments. **(1 mark)**

IV) What is the resultant torque about the point O? **(3 marks)**

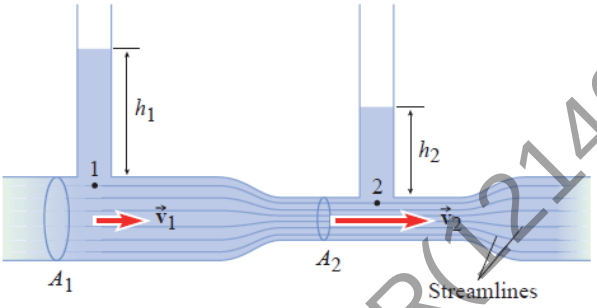
17) Venturi meter (Fig. below) measures fluid speed in a pipe. A constriction (of cross-sectional area A_2) is put in a pipe of normal cross-sectional area $\boldsymbol{A_1}$. Two vertical tubes, open to the atmosphere, rise from two points, one of which is in the constriction. The vertical tubes function like manometers, enabling the pressure to be determined. From this information the flow speed in the pipe can be determined. Suppose that the pipe in question carries water, $\boldsymbol{A_1 = 2A_2}$, and the fluid heights in the vertical tubes are h_1 = 1.20m and h_2 = 0.80 m.

a) Define venturi meter referred to fluid dynamics. **(1 mark)**

b) Find the ratio of the flow speeds v_2/v_1. **(2 marks)**

c) Find the gauge pressures P_1 and P_2. **(4 marks)**

d) Find the flow speed v_1 in the pipe. **(3 marks)**



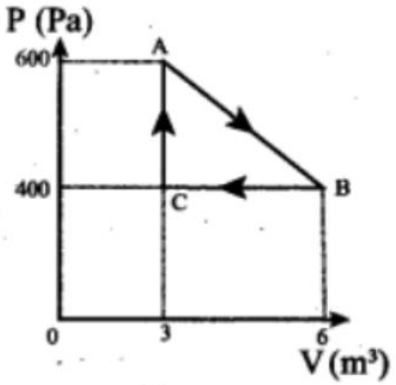
18) Based on principles of heat and thermodynamics answer the following questions:

I) Define:

a) specific heat capacity **(1 mark)**

b) heat capacity **(1 mark)**

II))An ideal gas is taken in a cyclic process as shown in the figure below:



In which process the work is done?

a) by a gas **(1 mark)**

b) on a gas **(1 mark)**

III) Referring to the diagram in (II) Calculate:

a) work done by the gas. **(2 marks)**

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b) work done on the gas **(2 marks)**
c) Net work done in the process. **(2 marks)**

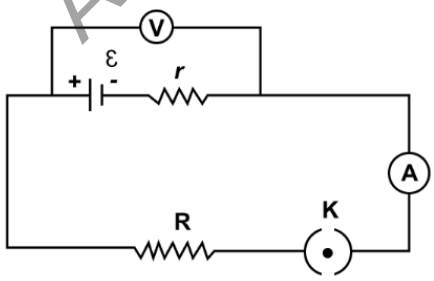
19) Here is given components of energy changes in a hydroelectric power station in disorder.

Mechanical energy of the turbine	Electrical energy	Gravitational potential energy of water in a dam or water fall	kinetic energy of falling water in the tunnel	mechanical energy in the dynamo

I) Give the correct sequence of energy changes in a hydroelectric power station, starting from the stored water. **(5 marks)**
II) State two advantages of hydropower. **(2 marks)**
III) Give a reasons why hydropower ultimately comes from the sun **(1 mark)**
IV) State two disadvantage of wind energy **(2 marks)**

20) A dry cell is a source of direct current (DC) electricity. It functions as an electrochemical cell that converts stored chemical energy into electrical energy. It provides a continuous, low-voltage flow of electricity, making it ideal for portable devices like flashlights, toys, and remote controls.
I) Differentiate emf of a cell and terminal voltage of a cell. **(2 marks)**
II) The diagram in figure below shows a cell of e.m.f. ε = 4V and internal resistance r = 2Ω connected to an external resistance R = 8Ω. The ammeter **A** measures the current in the circuit and the voltmeter **V** measures the terminal voltage across the cell. Find the readings of the ammeter and voltmeter when:
a) the key K is open **(4 marks)**
b) the key K is closed. **(4 marks)**

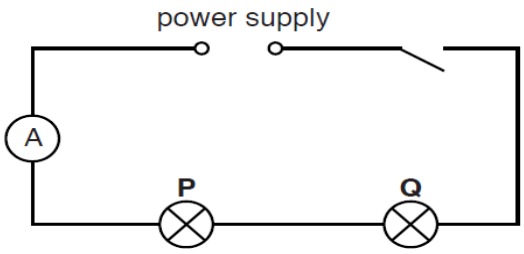
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SECTION C: COMPULSORY QUESTION (15 marks)

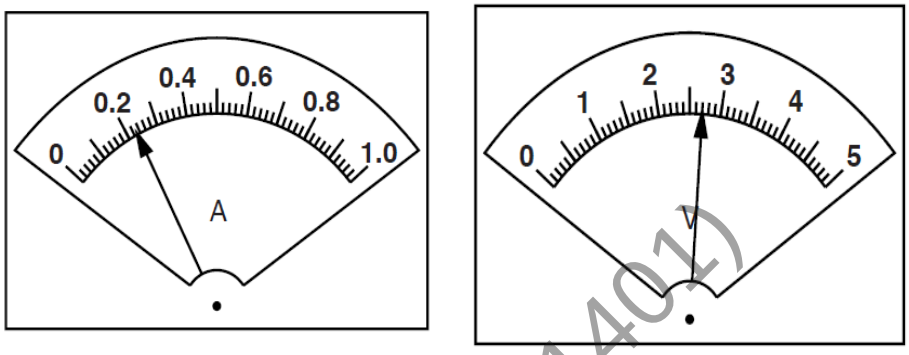
21) The class is investigating a circuit containing two lamps in series. They are using the circuit shown in Fig below:

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I) On Figure. above, use the standard symbol to show a voltmeter connected to measure the potential difference (p.d.) across lamp **P**. **(1 mark)**

II) Record the current *I* and the voltage **V** in the circuits, as shown on the ammeter and the voltmeter in Figure. below: **(2 marks)**



III) The crocodile clip is connected in turn to the resistance wire at points **A**, **B**, **C, D, E** and **F**. The potential difference ***V*** and current ***I*** are measured for each position and recorded in Table below:

position	V/V	I/A
A	1.3	0.20
B	1.2	0.35
C	1.1	0.46
D	0.9	0.74
E	0.8	0.87
F	0.6	1.13

a) Plot a graph of *V* / V (*y*-axis) against *I* / A (*x*-axis). **(8 marks)**

b) Determine the gradient **M** of the graph. Show clearly on the graph how you obtained the necessary information. **(2 marks)**

c) What this gradient represents? **(1 mark)**

d) Justify **(1 mark)**

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